

NOTES FOR REACTIONS

1. All loading conditions are examined and only maximum/minimum H or V and the corresponding H or V are reported.
2. Positive reactions are as shown in the sketch. Foundation loads are in opposite directions.
3. Bracing reactions are in the plane of the brace with the H pointing away from the braced bay. The vertical reaction is downward.
4. Building reactions are based on the following building data:

Width

(mm)

= 22596

Length

(mm)

= 22670

Eave Height

(mm)

= 7300 / 7300

Roof Slope

(rise/100)

= 25.00 / 25.00

Roof Dead Load

(kN/m2)

= 0.10

Wall Dead Load

(kN/m2)

= 0.10

Left Endwall

(kN/m2)

= 0.10

Right Endwall

(kN/m2)

= 0.10

Front Sidewall

(kN/m2)

= 0.10

Back Sidewall

(kN/m2)

= 0.10

Live Load

(kN/m2)

= 0.57

Collateral Load

(kN/m2)

= 0.05

Wind Speed

(m/s)

= 36.1

Wind Code

= MBMA10

Exposure

= B

Closure

= Enclosed

Internal Wind Coeff

= -0.18, +0.18

Risk Category

= -

Importance - Wind

= 1.00

Importance - Seismic

= 1.00

Seismic Design Category

= B

Seismic Coeff

(Sms)

= 0.33

Temperature Change

= 20

5. Loading conditions are:

1 Dead+Collateral+Live

2 Dead+Collateral+0.75Live+0.75Wind_Long2R+0.75Floor_Live+0.75LWIND2_L2E

3 Dead+Collateral+0.75Live+0.75Wind_Long2R+0.75Floor_Live+0.75LWIND2_R2E

4 0.6Dead+Wind_Left1

5 0.6Dead+Wind_Right1

6 0.6Dead+Wind_Left2

7 0.6Dead+Wind_Right2

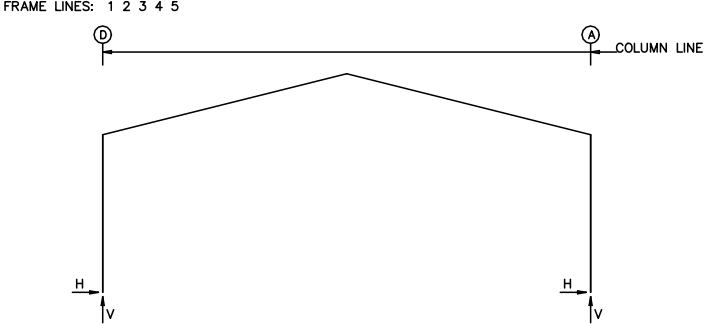
8 0.6Dead+Wind_Long1L+LWIND1_L2E

9 0.6Dead+Wind_Long1L+LWIND1_R2E

10 1.02Dead+1.02Collateral+0.52Seismic_Right+0.75Floor_Live

11 0.6Dead+Wind_Right2+Wind_Suction

12 0.6Dead+Wind_Pressure+Wind_Long2L



RIGID FRAME: ANCHOR BOLTS & BASE PLATES

Frm Line	Col Line	Anc._Bolt Qty	Dia	Base_Plate (mm)			Grout (mm)
				Width	Length	Thick	
1*	D	4	20	200	300	16	500
1*	A	4	20	200	300	16	500

1* Frame lines: 1 5

RIGID FRAME: ANCHOR BOLTS & BASE PLATES

Frm Line	Col Line	Anc._Bolt Qty	Dia	Base_Plate (mm)			Grout (mm)
				Width	Length	Thick	
2*	D	4	20	200	300	16	500
2*	A	4	20	200	300	16	500

2* Frame lines: 2 3 4

RIGID FRAME: BASIC COLUMN REACTIONS (kN)

Frame Line	Column Line	----Dead----		--Collateral--		----Live----		--Wind_Left1--		--Wind_Right1--		--Wind_Left2--	
		Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert
1*	D	3.0	8.0	0.8	1.7	9.0	18.3	-13.1	-18.9	-0.4	-12.2	-12.8	-13.5
1*	A	-3.0	8.0	-0.8	1.7	-9.0	18.3	0.4	-12.2	13.1	-18.9	0.1	-6.7

Frame Line	Column Line	--Wind_Right2--		--Wind_Long1--		--Wind_Long2--		--Seismic_Left		Seismic_Right		Seismic_Long1	
		Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert
1*	D	-0.1	-6.7	-3.0	-27.8	-2.3	-22.3	-0.9	-0.5	0.9	0.5	0.0	-2.0
1*	A	12.8	-13.5	3.0	-27.8	2.3	-22.3	-0.8	0.5	0.8	-0.5	0.0	-2.0

Frame Line	Column Line	Seismic_Long2		LWIND1_L2E--		LWIND1_R2E--		LWIND2_L2E--		LWIND2_R2E--	
		Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert
1*	D	0.0	2.0	0.1	-1.7	-0.4	-0.3	0.1	-1.7	-0.4	-0.3
1*	A	0.0	2.0	0.4	-0.3	-0.1	-1.7	0.4	-0.3	-0.1	-1.7

Frame Line	Column Line	----Dead----		--Collateral--		----Live----		--Wind_Left1--		--Wind_Right1--		--Wind_Left2--	
		Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert
2*	D	4.2	10.5	1.4	2.9	15.8	32.2	-18.6	-24.8	3.9	-15.0	-19.9	-15.1
2*	A	-4.2	10.5	-1.4	2.9	-15.8	32.2	-3.9	-15.0	18.6	-24.8	-2.7	-5.4

Frame Line	Column Line	--Wind_Right2--		--Wind_Long1--		--Wind_Long2--		--Seismic_Left		Seismic_Right		Seismic_Long1	
		Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert
2*	D	2.7	-5.4	-5.3	-37.9	-4.0	-28.2	-1.0	-0.6	1.0	0.6	0.0	-2.0
2*	A	19.9	-15.1	5.3	-37.9	4.0	-28.2	-1.0	0.6	1.0	-0.6	0.0	-2.0

Frame Line	Column Line	Seismic_Long2		LWIND1_L2E--		LWIND1_R2E--		LWIND2_L2E--		LWIND2_R2E--	
		Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert	Horz	Vert
2*	D	0.0	2.0	0.1	-3.0	-0.8	-0.6	0.1	-3.0	-0.8	-0.6
2*	A	0.0	2.0	0.8	-0.6	-0.1	-3.0	0.8	-0.6	-0.1	-3.0

1* Frame lines: 1 5

2* Frame lines: 2 3 4

ENDWALL COLUMN: BASIC COLUMN REACTIONS (kN)

Frm Line	Col Line	Dead Vert	Wind Press Horz	Wind Suct Horz	Seis Long Vert
1	C	1.4	-15.2	15.8	0.1
1	B	1.4	-15.2	15.8	0.1
5	B	1.4	-15.2	15.8	0.1
5	C	1.4	-15.2	15.8	0.1

ENDWALL COLUMN: ANCHOR BOLTS & BASE PLATES

Frm Line	Col Line	Anc._Bolt Qty	Dia	Base_Plate (mm)			Grout (mm)
				Width	Length	Thick	
1	C	4	20	200	300	16	500
1	B	4	20	200	300	16	500
5	B	4	20	200	300	16	500
5	C	4	20	200	300	16	500

BUILDING BRACING REACTIONS

---Wall Loc	---Col Line	Reactions in plane of wall (kN)				of wall Panel_Shear (N/m)		Note
		Wind	Seismic	Wind	Seis	Wind	Seis	
L_EW	1							(h)
F_SW	A	1,2	*	11.8	2.3			
		4,5	*	11.8	2.3			
R_EW	5							(h)
B_SW	D	5,4	*	11.8	2.3			
		2,1	*	11.8	2.3			

(h)Rigid frame at endwall

*See RF reactions table for vertical and horizontal reactions in plane of the rigid frame.

Reactions for seismic represent shear force, Eh
Reaction values shown are unfactored

Sivoko Steel Structures Ltd.

PROJECT	Proposed Warehouse	ANCHOR BOLT REACTIONS		
ID	Q-1467	DESIGN:	DRAFT:	CHECK:
PROJECT ADDRESS	Proposed Warehouse	DATE: 15/DEC/2025\$ SHEET 5 OF 5		